

Problem set 4
Due Saturday February 19 at 11:59pm

Exercise 4.1 (10 points; part of Rudin 2.9). Let E be a subset of a metric space. Recall that the *interior* of E , denoted E° , is the set of all interior points of E .

- (1) Prove that E° is always open.
- (2) Prove that E is open if and only if $E^\circ = E$.
- (3) Prove that, if G is an open subset of E , then $G \subset E^\circ$.

Exercise 4.2 (10 points). Show that the union of a finite number of compact subsets of a metric space X is compact.

Exercise 4.3 (10 points; Rudin 2.25, modified). A metric space which has a dense subset which is at most countable is called *separable*. For example, $\mathbb{Q}$ is a countable, dense subset of $\mathbb{R}$, so $\mathbb{R}$ is separable.

- (1) Suppose that K is a compact metric space. Prove that K is separable. (Hint: first show that for every $n \in \mathbb{N}$, there are finitely many neighborhoods of radius $1/n$ whose union covers K .)
- (2) Give an example of a metric space which is not separable.

Exercise 4.4 (10 points; Rudin 2.19, modified). We say a metric space X is *connected* if it cannot be written as a disjoint union of two nonempty open sets. (This definition is slightly different from that in Rudin; you can show it is equivalent.)

- (1) Show that X is connected if and only if the only clopen (closed and open) subsets of X are $\emptyset$ and X .
- (2) Fix $p \in X$ and $\delta > 0$. Define $A = \{q \in X \mid d(p, q) < \delta\}$. Define $B = \{q \in X \mid d(p, q) > \delta\}$. Show that A and B are open sets.
- (3) Prove that every connected metric space with at least two points is uncountable. (Hint: use (2))